

A Hybrid VR/Desktop Sculpting with Substance Modeler

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1 INTRODUCTION

Substance 3D Modeler is the first digital shaping tool that is built from the ground up with a unified spatial and desktop workflow and architecture. The UX is designed to encourage creative flow as much as possible, allowing artists to have a tactile and hand-crafted sculpting experience on the computer while also harnessing flexible tools and powerful scene assembly features. The signed distance field (SDF) sculpting engine is built with responsiveness in mind, and to enable a clay-like feel when designing forms. Artists can focus as much as possible on the creative process without letting topology and other technical constraints get in the way.

The advantages of our approach are:

- The prioritization of *creative flow*, from our UI architecture and design down to our surface tech and performance / responsiveness.
- The UI and rendering is unified between desktop and VR. Artists can choose whichever surface works best for their shaping workflow.
- Artists do not have to worry about mesh topology or other technical constraints and can focus more fully on the creative process.

2 A BRIDGE BETWEEN SPATIAL AND DESKTOP COMPUTING

Reducing the friction between working in VR and integrating it within existing professional desktop workflows has been one of the main challenges of spatial hardware. One of our firm beliefs about spatial computing tools is that as a prerequisite for them to add substantive value for modern productivity use cases, they must be thoroughly integrated into prior-existing professional computing workflows. This is especially so for content creators and developers who intend to build experiences for the next generation of hardware or 3D content targets.

The 6-DOF input and display capabilities of contemporary VR offer users a more direct experience when compared to traditional

desktop tools, which enable artists to craft complex digital 3D content and scenes that more naturally make use of human sensory input. The artist can easily translate their hand motions into creative decisions, while also viewing the digital 3D scene without any distortion and with a sense of real-world scale and presence.

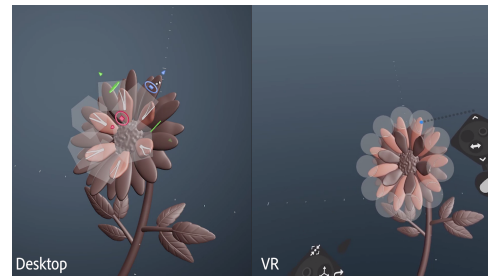


Figure 1: Using the warp sculpting tool on both desktop (left) and VR (right)

With Modeler, these spatial capabilities are added to existing desktop workflows without abandoning the more traditional display and inputs (desktop mouse and tablet). We consider spatial and traditional digital input & displays as partners with distinct benefits given the artist's given task, and so Modeler is designed with an additive approach to inputs & display experiences. The goal of this tool is to allow "shaping on all surfaces", to enable the artist to use whichever inputs and whatever display surfaces that best suit their needs.

3 CLAY-INSPIRED SCULPTING

We wanted the sculpting experience in Modeler to feel like the volumetric workflow of real clay. Signed distance fields were chosen for the sculpting engine so that artists could create and modify topology and as the artist edits the SDF, a triangle mesh is generated from the SDF for rendering. The triangle mesh is cached on the GPU where it is then rasterized every frame using a compute shader based, GPU-driven rendering pipeline. Editing operations happen asynchronously with rendering images for the VR headset or desktop monitor by using multiple CPU threads and GPU contexts for editing and rendering. The SDFs are stored as any number of independent 3D voxel grids, each with a different 3D transform associated with them, with a typical sculpt consisting of a thousands of grids. This concept of multiple, high resolution SDF voxel grids enables Modeler to use SDFs without sacrificing (much) resolution; one grid can capture detail at sub-millimeter resolution while other grids capture lower resolution detail.

A distinction between Modeler's sculpting engine and other volumetric implementations is the use of copy-on-write (COW) semantics throughout the application's data structures. With COW,

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Figure 2: An organic sculpt created in Desktop and VR by Giovanni Nakpil

multiple voxel grids can share data with each other which means that instancing of a voxel grid and subsequent edits to part of a copied grid are nearly instantaneous, and cheap snapshots are used to support undo/redo, automatic background saves, and other useful shaping features. While other volumetric databases exist, none support COW across multiple high resolution voxel grids.

On the sculpting tool side, Modeler uses analytical signed distance functions as tool shapes which are customizable by the artist. These shapes affect the underlying voxel grid using CSG operations. Modeler also has different tools which use either analytic SDFs or triangle meshes to edit discretized SDF grids. The ability to fluidly change between surface representations is a core concept of Modeler's sculpting engine, and while these changes are not lossless, the discretized SDF grids are at sufficiently high resolution to minimize lossiness, made possible with modern computing power.

4 KITBASH-INSPIRED SCENE ASSEMBLY

The inspiration of Modeler's scene assembly workflow comes from physical model kitbashing. Our assembly tools allow the artist to quickly arrange the voxel layers both in space and in grouping hierarchy. When the artist repeats objects across a scene, the copy of a node and visuals happens instantaneously, while the memory is silently shared under the hood until the artist modifies a copy with a sculpting tool (using the COW mechanism described earlier). This gives the artist the illusion of working with physical copies of the data, without bogging them down with worrying about memory constraints. While we still present the artist with an option of instancing layers and groups, we do so purely to let them keep copies synchronized, rather than as a memory management method. We also provide tools to rapidly make array, mirror, and radial copies of the data for common symmetric structures found in mechanical and architectural objects.

This artistic freedom coupled with rapid repetition tools can easily produce vast quantities of scene nodes, up to the hundreds of thousands. To maintain the illusion of a complex scene with large numbers of objects, we rely on a heavily optimized scene graph structure which can be updated in a parallel and cache friendly way. This rapid node proliferation also makes traditional list views of the scene hierarchy hard to navigate, so we take the unconventional approach of doing all scene management directly in the viewport,

rather than indirectly through a hierarchical list of node names. This approach is also congruent with our ethos of working on the scene directly, in essence applying WYSIWYG principles to 3D creation. Interestingly, we have found that when we present artists with an actual text-based list of their scene structure, they become more timid with their editing. We hypothesize that seeing the listing produces a subconscious desire to keep it "usable" and thus leads to self-imposed limits on creative output.

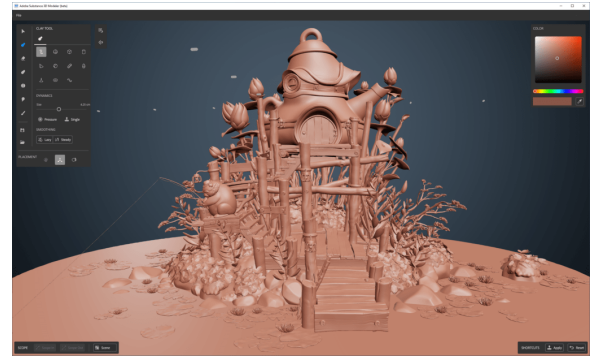


Figure 3: A complex scene that used various assembly features, by Joshua Eiten

While we strive to present a nontechnical way of working with the scene, Modeler still provides a robust set of operations that are more native to a technical environment. For example, we support procedural mirror and radial repetition of sub-graphs, so it is easy to make symmetric structures in a non-destructively. We also allow instanced copying of sub-graphs, to easily re-use parts of your project in a repeated fashion. Edit any instance and the changes propagate to all others. At any moment the artist can "break" instancing to targeted content, in order to add unique detail to parts of their creation. Lastly, we support recursion within the graph by the deceptively simple act of adding an instance of a group to itself.

In combination, these features allow new creative paradigms and workflows to emerge that feel more intuitive and fun than more technical methods of construction. For example, when copying is cheap both in time and memory, it becomes intuitive, fun, and fast to build a tree out of individual branches and leaves. This lets us bypass teaching artists more technical methods of construction and instead assemble them intuitively "as things are", while still reaping the digital benefits of replication when desired.